Manugunta Meghana Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.10
FIITJEE	Class XII (Telangana State Board of Intermediate Education)	2023	98.10%
Silver Oaks High School	Class X (CBSE)	2021	98.40%

SCHOLASTIC ACHIEVEMENTS

Selected for the Sakura Science Program to do research on **photonic beam steering** at the **University of Tokyo.**

NTSE Scholar : Recognized as one of the **811 scholars** from a pool of **~1,000,000** applicants at **national level**.

EXPERIENCE AND POSITIONS OF RESPONSIBILITY

Semester Internship at Texas Instruments

- Engineered and validated a **high-voltage test PCB** for **Safe Operating Area (SOA)** characterization of **GaNFET** power transistors.
- Debugged an existing **clamp circuit** and designed a **synchronous circuit** for **dRon characterization** across voltage, frequency, and temperature in an **industrial HV lab**.

Teaching Assistant - C Programming

Teaching Assistant - Waveguides and Antennas

Core in Elektronica (Electronics club of IIT H)

- Organized and coordinated technical sessions and hands-on workshops for students.
- Worked on electronics projects involving **sensors, Arduino, Raspberry Pi** and **hardware interfacing**.

SKILLS

Tools : KICAD | LTspice | Altium | Pspice | Cadence Virtuoso | Ques | Arduino IDE | Solid Edge (CAD) | Ngspice

FPGA & Microcontroller Boards : Vaman (FPGA) | Arduino UNO | Raspberry Pi

Programming Languages : Verilog | C | C++ | Python | MATLAB | Bash | Git | SQL | LaTeX

PROJECTS

Balanced Microring True-Time-Delay Beamforming | Photonics

- Designed and optimized **microring-based true-time-delay (TTD) elements** for wideband beam steering, analyzing group delay, delay-bandwidth product, insertion loss, and amplitude ripple across the operating frequency range.
- Developed MATLAB models for **single and cascaded microring configurations**, optimizing ring radius and coupling coefficient to obtain improved delay characteristics and evaluating their impact on wideband phased-array beam steering.
- Extended the analysis toward **far-field wideband beam steering**, investigating the role of true-time delay in mitigating **beam squint** across frequency.

Wideband Switched-Type Attenuator

- Designed and implemented a **DC-50 GHz switched-type attenuator (STA)** using hybrid **Simplified-T, T-type, and -type attenuation cells** to realize programmable attenuation steps from **0.5-8 dB**.
- Derived **S-parameter-based** design equations for **impedance matching** and **attenuation**, and developed **numerical optimization routines in C** to determine the required series and shunt resistance values.
- Validated **32 attenuation states in Cadence**, achieving **16.5 dB total attenuation range, 0.38 dB RMS amplitude error** and **3.17° RMS phase error.**

RISC-V pipelined processor

- Designed and implemented a **pipelined RISC-V processor** in **Verilog**, developing the datapath, pipeline stages, and control logic for instruction execution.
- Implemented comprehensive **pipeline hazard handling**, including data/control hazard detection, **stall mechanisms**, and **branch instruction** handling to maintain correct instruction flow.

LDO Design | Cadence, 180-nm CMOS

- Designed and simulated a **capacitorless low-dropout regulator (LDO)** with an error amplifier (EA) targeting high loop gain and stability without requiring a large external output capacitor.
- Designed and optimized the EA to achieve **63 dB DC gain and 66° phase margin**, analyzing loop stability and frequency response through Cadence simulations.
- Incorporated a **flipped-voltage-follower (FVF) stage** for fast load-transient response.

RTL Design of Fixed-Point Softmax Unit

- Architected and implemented a fully synthesizable, pipelined Softmax hardware unit in Verilog.
- Developed hardware-friendly **exponential** and **logarithmic** approximations using **range-reduction techniques.**
- Created **Verilog testbenches** to validate **functional correctness**, comparing RTL outputs against expected Softmax results and debugging timing, intermediate values, and numerical behavior using **GTKWave**.

Adaptive Noise Cancellation Headphones

- Developed a real-time adaptive noise-cancellation system using **RLMS-based full suppression** to estimate and cancel unwanted noise.
- Implemented a **partial-suppression mode** by using **NLMS** to estimate user-selected tonal frequencies and combining the estimated tonal component with the RLMS full-suppression output, enabling **selective preservation** of desired tones.